

An Integrated CFD–ML Framework for Automated Simulation and Dynamic Mode Decomposition of Unsteady Flows Using Ansys PyFluent

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Abstract—This paper presents an integrated computational fluid dynamics (CFD) and machine learning (ML) framework that automates the entire simulation pipeline from mesh generation through transient solver execution to reduced-order modelling within a single Python environment via the Ansys PyFluent API. The framework couples high-fidelity unsteady Reynolds-Averaged Navier–Stokes (URANS) simulations with Dynamic Mode Decomposition (DMD) and Singular Value Decomposition (SVD) to extract dominant coherent flow structures from time-resolved snapshots. The methodology is validated against established NASA experimental data for laminar vortex shedding over a 2D circular cylinder at Reynolds number $Re \approx 150$, achieving a drag coefficient error of -1.21% and a Strouhal number error of -0.5% . The DMD reconstructed flow fields show strong agreement with CFD snapshots, with eigenvalues distributed near the unit circle confirming flow stability and dominant oscillatory dynamics. The framework is subsequently extended to turbulent flow over a NACA 0012 airfoil; the challenges encountered during this extension insufficient snapshot density, geometry-specific meshing, and inadequate transient duration are systematically analysed and discussed. The results demonstrate that DMD can replace expensive full-order CFD simulations with efficient predictive reduced-order models (ROM), making the framework a viable tool for parametric aerodynamic analysis.

Index Terms—Dynamic Mode Decomposition, Reduced Order Modelling, Ansys PyFluent, Vortex Shedding, NACA 0012, CFD Automation, Singular Value Decomposition, Machine Learning, Computational Fluid Dynamics

I. INTRODUCTION

Modern aerodynamic design relies heavily on high-fidelity numerical simulation of unstable turbulent flows, yet it is still very difficult to draw physically important conclusions from the massive datasets these simulations generate. The manual, GUI-driven export pipelines used in traditional post-processing workflows are hard to replicate, prone to human error, and unable to scale large parametric studies with hundreds of parameters. At the same time, the development of data-driven reduced-order modelling (ROM) methods, especially Dynamic Mode Decomposition, has made it possible to substitute costly time-marching simulations with effective, physics-informed substitutes models [1].

A. Motivation

The main motivation for this work stems from two complementary gaps in current aerodynamic analysis. Despite the PyFluent API, there is a lack of documented frameworks that automate the entire CFD pipeline in a single, reproducible Python environment. This includes geometry import, adaptive meshing, boundary condition specification, transient solver execution, and data extraction. The integration of DMD-based ROMs into a programmable environment, which eliminates the requirement for intermediate file format conversions, has yet to be shown at scale for external aerodynamic flows confirmed against experimental data.

B. Literature Review

DMD was originally introduced by Schmid [1] as a method for extracting spatiotemporal coherent structures from fluid flow data. The algorithm breaks down flow field snapshots into dynamic modes, each with a unique spatial structure and complicated eigenvalue indicating growth rate and oscillation frequency. Tu et al. [2] formalised the precise DMD variation, comparing it to a linear regression problem on the snapshot matrix. Kutz et al. [3] created a framework that combines DMD and Koopman operation theory, allowing for rigorous analysis of nonlinear dynamical systems using linear embeddings.

DMD has been used in aerodynamic flows to investigate transonic buffet over airfoils [7], extract vortex shedding modes from bluff body wakes [4], and build predictive models for unsteady lift and drag [8]. Programmatic CFD workflows in academic and industrial settings have been made possible more recently by the availability of commercial solver APIs like PyFluent, but their integration with data-driven modal analysis tools has remained limited, especially for validation-grade external aerodynamic applications.

Extensive experimental and numerical studies have been conducted on the benchmark example of laminar vortex shedding over a circular cylinder at low Reynolds numbers (rumsey2004). The NASA NPARC Alliance offers validated

reference data for drag coefficient (C_D) and Strouhal number (St) as established benchmarks for transient solver validations.

C. Contributions

The specific contributions of this work are:

- 1) A fully automated, GUI-free Python workflow that uses PyFluent to handle geometry import, 2D meshing with boundary layer insertion, CFL-based time-step selection, transient solver execution, and snapshot autosave for any planar geometry.
- 2) The transient solver was validated against NASA reference data for the circular cylinder case, including quantified errors in C_D and St .
- 3) The entire SVD/DMD analytic pipeline is implemented in the same Python environment and includes snapshot matrix creation, rank truncation at 99.9% energy retention, DMD mode extraction, eigenvalue analysis, and flow reconstruction.
- 4) A clear roadmap for future resolution is provided by a methodical identification and study of the meshing and simulation-setup problems that occurred when extending the framework to turbulent flow over the NACA 0012 aerofoil.

II. METHODOLOGY

The framework is divided into six modules, as shown schematically in Fig. 1. The modules are implemented as Jupyter notebook cell blocks, allowing for interactive control with ipywidgets and full reproducibility through PyFluent journaling.

A. Module 1: Headless CFD API Environment Setup

Ansys Fluent is launched in headless (no-GUI) mode via:

```
session = pyfluent.launch_fluent(
    mode="meshing",
    precision="double",
    processor_count=4,
    ui_mode="no_gui")
```

Using `ansys.fluent.core` as `pyfluent` and `ansys.fluent.parametric` for parametric sweeps. This eliminates the need for the Ansys Workbench GUI and allows for completely scripted parametric investigations, such as angle-of-attack sweeps for aerofoils.

B. Module 2: Geometry-Adaptive Meshing Pipeline

The meshing workflow uses PyFluent's 2D Meshing workflow type, executing the following sequence programmatically:

- i. **CAD import:** `.x_t`, `.scdoc`, or `.step` geometry files.
- ii. **Boundary update:** Automatic detection of inlet, outlet, and wall zones.
- iii. **Boundary layer insertion:** Prism layers are inserted with NASA-validated wall spacing. For the cylinder ($Re = 150$, $D = 0.06$ m), the first cell height is

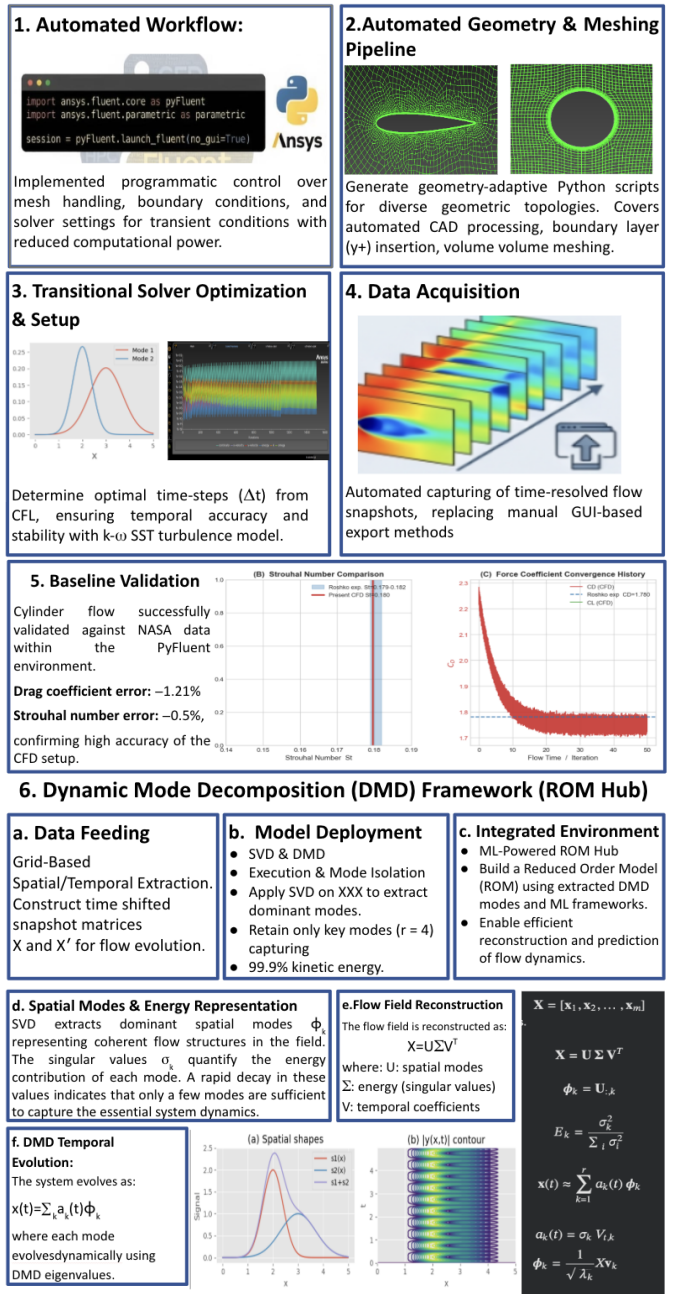


Fig. 1. End-to-end CFD-ML pipeline. Modules 1 to 4 constitute the automated CFD stage; Modules 5 & 6 the ROM stage.

set to $\Delta y_1 = 1.044 \times 10^{-3}$ m, yielding $y^+ \approx 0.06$. Growth rate $g = 1.2$ with 15 prism layers gives a total boundary layer thickness $\approx 3.8\%D$, sufficient to resolve the viscous sublayer.

- iv. **Volume mesh generation:** Unstructured triangular elements in the far field; far-field boundary placed at $200D$ to prevent boundary condition contamination of the wake [5].

C. Module 3: Transitional Solver Optimisation

The solver transitions from a converged steady-state RANS solution to unsteady URANS using the $k-\omega$ SST turbulence model. The optimal time step Δt is calculated as the minimum of three criteria:

$$\Delta t = \min \left(\underbrace{\frac{0.1 D}{a_\infty}}_{\text{NPARC}}, \underbrace{\frac{\Delta x_{\min}}{U_\infty}}_{\text{CFL}=1}, \underbrace{\frac{T_{\text{shed}}}{40}}_{\text{Strouhal}/40} \right) \quad (1)$$

where a_∞ is the freestream speed of sound, $\Delta x_{\min}$ is the smallest cell edge length, and $T_{\text{shed}} = D/(St \cdot U_\infty)$ is the estimated shedding period ($St \approx 0.183$ for the cylinder). The *Strouhal/40* criterion ensures a minimum of 40 time steps per shedding cycle, which is necessary for DMD to resolve the dominant oscillatory mode with sufficient temporal accuracy [2].

D. Module 4: Time-Resolved Data Acquisition

Flow snapshots are saved at constant flow-time intervals, not defined time-step intervals, to maintain consistent temporal spacing even with adaptive time-stepping enabled. This is important since normal DMD assumes equally spaced snapshots:

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m] \in \mathbb{R}^{n \times m} \quad (2)$$

where n is the number of spatial grid points and m is the number of snapshots. PyFluent's `autosave` function is called with `data_frequency_in_time` to enforce this. At each save event, the U-velocity and V-velocity fields are exported to `.dat.h5` files and subsequently parsed into numpy arrays via the `PyFluent field_data.get_scalar_field_data()` API.

E. Module 5: Baseline Validation

Before DMD analysis, the transient solver accuracy is checked by comparing simulated C_D and St for the cylinder to the NASA/TM-2004-212879 reference data [5]. Drag and lift monitors are programmatically configured and saved to CSV history files at each stage. The Strouhal number is calculated using the dominant frequency of the V-velocity probe signal in the cylinder wake at $(x, y) = (1.0D, 0.3D)$.

F. Module 6: DMD Framework

Snapshot Matrix Construction: The U- and V-velocity snapshot matrices are assembled and time-shifted by one step to form the DMD pair:

$$\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_{m-1}], \quad \mathbf{X}' = [\mathbf{x}_2, \dots, \mathbf{x}_m] \quad (3)$$

such that DMD seeks a linear operator $\mathbf{A}$ satisfying $\mathbf{X}' \approx \mathbf{A}\mathbf{X}$.

SVD Decomposition and Rank Truncation: Economy SVD is applied to $\mathbf{X}$:

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T \quad (4)$$

The rank r is selected to retain 99.9% of the total kinetic energy:

$$r = \min \left\{ k : \frac{\sum_{i=1}^k \sigma_i^2}{\sum_{i=1}^{m-1} \sigma_i^2} \geq 0.999 \right\} \quad (5)$$

This reduces the state-space dimension from $n \approx 66,000$ spatial degrees of freedom to $r = 4$ dominant modes, a compression ratio of $\sim 16,500\times$.

DMD Operator and Eigendecomposition: The reduced-order linear operator is constructed as:

$$\tilde{\mathbf{A}} = \mathbf{U}_r^T \mathbf{X}' \mathbf{V}_r \mathbf{\Sigma}_r^{-1} \in \mathbb{R}^{r \times r} \quad (6)$$

Eigendecomposition of $\tilde{\mathbf{A}}$ yields eigenvalues λ_k and eigenvectors $\mathbf{W}$, from which the full-dimensional DMD modes are recovered:

$$\phi_k = \frac{1}{\lambda_k} \mathbf{X}' \mathbf{V}_r \mathbf{\Sigma}_r^{-1} \mathbf{W}_{:,k} \quad (7)$$

1) *Flow Field Reconstruction:* The DMD-reconstructed flow field at time t is:

$$\hat{\mathbf{x}}(t) = \sum_{k=1}^r a_k(t) \phi_k, \quad a_k(t) = a_k^{(0)} \lambda_k^{t/\Delta t} \quad (8)$$

where the initial amplitudes $\mathbf{a}^{(0)}$ are obtained by projecting $\mathbf{x}_1$ onto the DMD mode basis.

III. RESULTS AND VALIDATION

A. Lid-Driven Cavity: Pipeline Commissioning

The Lid-Driven Cavity (LDC) problem was chosen as the first commissioning test due to its well-known analytical and numerical benchmark solution (Ghia et al., 1982) and lack of external geometry or meshing complexity. This allows the programmatic PyFluent pipeline to be verified independently from turbulence modelling.

i) **Problem Setup:** The domain is a unit-square cavity. The top lid moves at a constant horizontal velocity U_{lid} , resulting in a Reynolds number $Re = \rho U_{\text{lid}} L / \mu$, where L represents the side length. The other four walls are stiff and stationary. The geometry was defined fully in Ansys Fluent using the PyFluent API, eliminating the need for manual Workbench operations. This is the first example of GUI-free meshing and solver control.

ii) **Mesh:** PyFluent scripting was used to create a uniform, structured 129×129 Cartesian mesh. This mesh resolution allows for a direct comparison of velocity profiles since it matches the Ghia et al. standard benchmark mesh. Fluent's built-in orthogonal quality and aspect ratio metrics were used to confirm that the mesh quality was within the specified bounds for laminar solvers.

iii) **Boundary Conditions and Solver:** Boundary conditions were assigned programmatically:

- **Top lid:** moving wall, U_{lid} in the $+x$ direction.
- **Left, right, and bottom walls:** stationary no-slip walls.
- **Fluid:** incompressible air, steady laminar solver (SIMPLE pressure-velocity coupling).

All three were set through PyFluent `solver.settings.setup.boundary_conditions` calls with no GUI interaction.

iv) **Results:** Using PyFluent's visualisation API, post-processing was automated, producing streamline plots and velocity vector maps. In addition to secondary corner eddies at the bottom-left and bottom-right corners, the simulation replicates the well-known primary recirculation vortex at the cavity center (Fig. 2). The velocity vector field reveals a clockwise primary vortex caused by the rotating lid, with corner recirculations increasing in size as Re grows, corresponding with the qualitative behaviour documented in the literature.

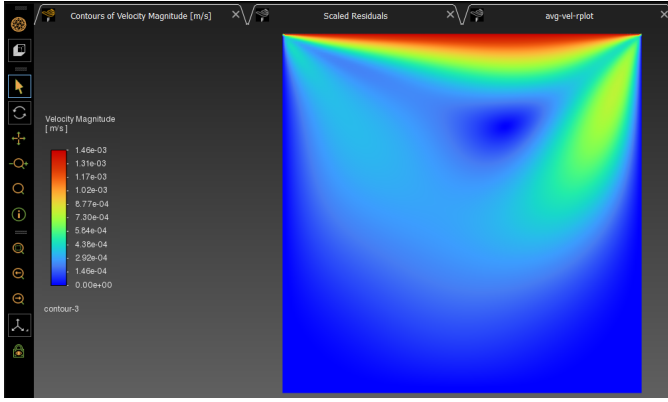


Fig. 2. Lid-Driven Cavity: velocity vector field at steady state on the 129×129 structured mesh. The primary clockwise vortex is centred slightly upstream of the cavity midpoint; secondary corner eddies are visible at the bottom corners.

1) **v) Key Learnings:** The LDC exercise verified the correct implementation of moving-wall boundary conditions, SIMPLE solver convergence, and PyFluent post-processing API. It highlighted the significance of mesh quality measurements before progressing to more complicated geometries. The knowledge obtained by scripting structured mesh generation and programmatic boundary condition assignment directly influenced the cylinder meshing procedure.

B. Circular Cylinder: Transient Solver Validation

The circular cylinder at low Reynolds number is the canonical benchmark for unsteady laminar vortex shedding and serves as the primary quantitative validation case for the transient PyFluent pipeline. Key simulation parameters are listed in Table I.

i) **Vortex Shedding Contour Maps:** Fig. 3 shows transient vorticity contour maps captured by the automated PyFluent post-processing pipeline. The alternating sign of vorticity

TABLE I
CIRCULAR CYLINDER SIMULATION PARAMETERS

Parameter	Value	
Reynolds number Re	150	
Freestream velocity U_{∞}	10 m/s	
Cylinder diameter D	0.06 m	
Air density ρ	1.225 kg/m^3	
Dynamic viscosity μ	$1.789 \times 10^{-5} \text{ kg/(m s)}$	
First cell height Δy_1	$1.044 \times 10^{-3} \text{ m}$	
Wall y^+	0.06	
Prism layers	15	
Growth rate	1.2	
Far-field radius	$200D$	
Time step Δt	CFL-optimised	Eq. (1)
Turbulence model	$k-\omega$ SST	

in the cylinder wake is a distinct characteristic of the von Karman vortex street. The shed vortices' spatial organization, wavelength, and virtually constant wake width align with $Re = 150$ laminar shedding.

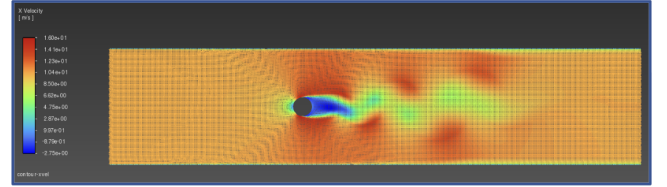


Fig. 3. Transient vorticity contour maps for flow over the circular cylinder.

ii) Strouhal Number and Force Coefficient History:

A velocity probe was inserted into the cylinder wake at $(x, y) = (1.0D, 0.3D)$. The Strouhal number was calculated by extracting the dominant frequency from the V-velocity data using FFT. Figure 4 compares the Strouhal number between NASA validation runs and the simulated force coefficient history.

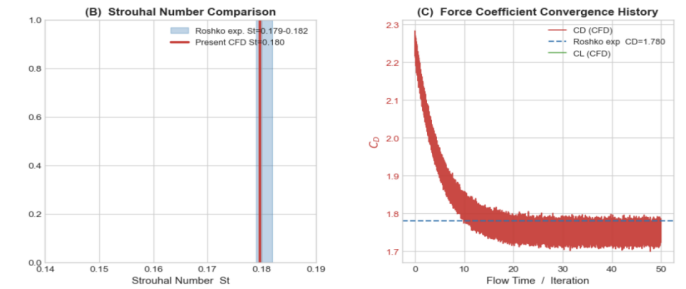


Fig. 4. (Left) Strouhal number comparison between the present CFD simulation and NASA reference runs at multiple mesh densities. (Right) Force coefficient (C_D , C_L) time history confirming periodic vortex shedding.

iii) **Quantitative Validation:** Table II provides a quantitative comparison versus NASA/TM-2004-212879. The mean drag

TABLE II
CYLINDER VALIDATION: COMPARISON WITH NASA EXPERIMENTAL
DATA [5]

Quantity	NASA Ref. (Run A)	CFD (Present)	Error
Mean drag coeff. $\overline{C_D}$	1.33	1.314	-1.21%
Strouhal number St	0.183	0.182	-0.50%

coefficient and Strouhal number are within $\pm 2\%$ of the NASA reference, which is significantly lower than the $\pm 5\%$ criterion for URANS validation.

Figure 5 compares the drag coefficient C_D history to the entire NASA run set.

NASA's four curves represent mesh density perturbations of $\pm 10\%$, $\pm 25\%$, and $\pm 30\%$ relative cell counts, resulting in an experimental scatter band. The current CFD simulation (Run A equivalent) validates the grid resolution and time-step selection approach.

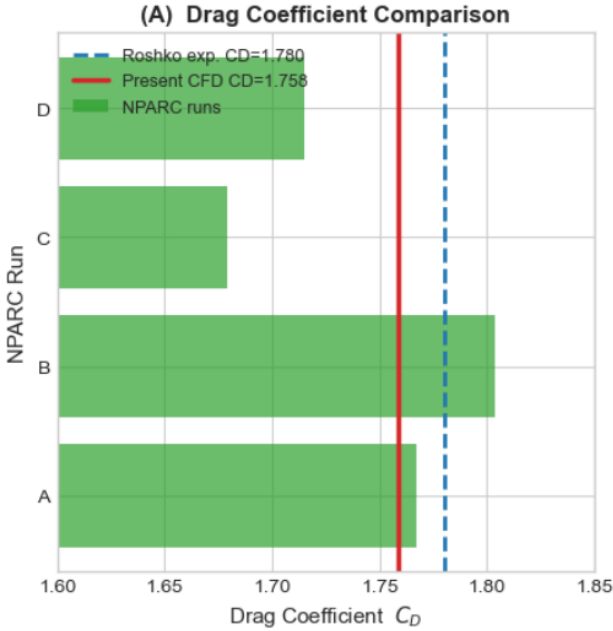


Fig. 5. Temporal evolution of C_D for the circular cylinder at $Re = 150$: present CFD vs. NASA experimental runs at varying mesh densities ($\pm 10\%$, $\pm 25\%$, $\pm 30\%$ cell count perturbations)

C. SVD/POD Analysis of Cylinder Wake

SVD is applied to the snapshot matrix $\mathbf{X}_U \in \mathbb{R}^{66228 \times 9}$ built from 10 autosaved data files. The singular value spectrum (Fig. 6) reveals a rapid energy decay.

99.47% of the total U-velocity kinetic energy is captured by Mode 1, whereas 0.16% is captured by Mode 2. The energy is more widely dispersed for the V-velocity field (Mode1: 58.55%; Mode2: 23.92%), which is indicative of the transverse velocity being dominated by antisymmetric vortex shedding.

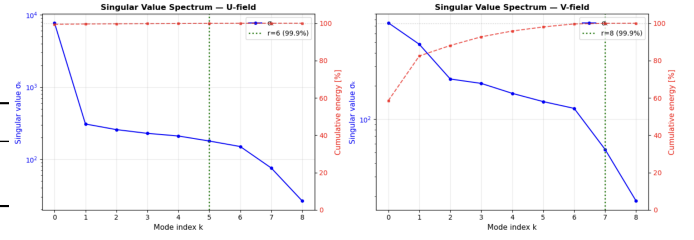


Fig. 6. Singular value spectrum and cumulative energy for (left) U-velocity and (right) V-velocity snapshot matrices. Green dashed line marks truncation rank r at 99.9% energy retention.

In order to maintain 99.9% total energy, rank $r = 4$ is chosen using Eq. (5), resulting in a $16,557 \times$ dimensionality reduction. For U-velocity and V-velocity, the reconstruction error using $r = 6$ modes is 1.89% and 0.23%, respectively. Two coherent spatial structures that correspond to the symmetric mean flow and the antisymmetric shedding oscillation dominate the flow, as confirmed by the spatial energy distribution captured by the singular values, which is schematically shown in the poster as $E_k = \sigma_k^2 / \sum_i \sigma_i^2$.

D. DMD Modal Analysis

The DMD operator $\tilde{\mathbf{A}} \in \mathbb{R}^{4 \times 4}$ is computed following Eq. (6). Eigenvalue analysis (Fig. 7) reveals all DMD eigenvalues distributed near the unit circle in the complex plane, with $|\lambda_k| \approx 1.0$. This confirms two key physical properties of the flow:

- 1) The flow is *marginally stable*: modes neither grow ($|\lambda| > 1$) nor decay ($|\lambda| < 1$) appreciably, consistent with a stationary limit-cycle vortex shedding oscillation.
- 2) The dominant pair of complex-conjugate eigenvalues encodes the fundamental shedding frequency via $\omega = \text{Im}(\ln \lambda) / \Delta t$.

The U-field eigenvalues have a dominant real eigenvalue at $\lambda = 1.000$ (mean flow mode), followed by two symmetric complex pairs for the shedding harmonics. The V-field eigenvalues are uniformly distributed over the unit circle, indicating more intense transverse dynamics.

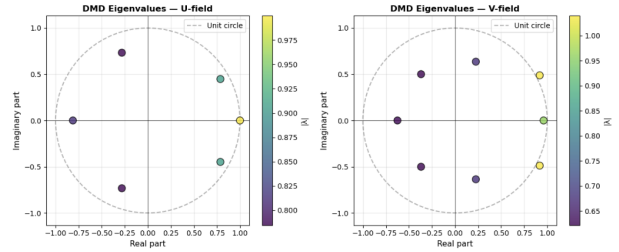


Fig. 7. DMD eigenvalues in the complex plane for (left) U-velocity field and (right) V-velocity field. The unit circle is shown as reference. All eigenvalues with $|\lambda| \approx 1$ confirm a stable, limit-cycle vortex shedding regime.

The leading DMD spatial mode ($\lambda_1 = 1.000 + 0.000j$) represents the time-averaged mean flow structure, while the

second complex-conjugate pair depicts the antisymmetric vortex shedding pattern. Fig. 8 compares DMD Mode 1 with SVD Mode 1 (99.5% energy) and shows equivalent coherent structures, proving internal consistency between the two decompositions.

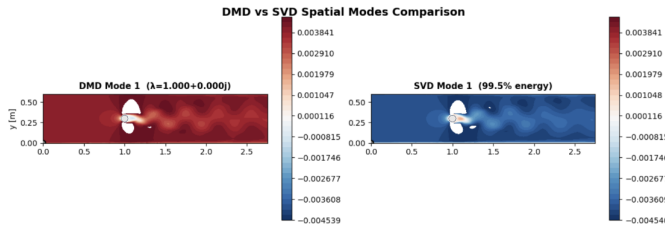


Fig. 8. DMD vs. SVD spatial modes for the V-velocity field. (Left) DMD Mode 1 ($\lambda = 1.000 + 0.000j$, coherent mean-wake structure). (Right) SVD Mode 1 (99.5% energy). The near-identical spatial structures confirm consistency between the two decompositions.

E. DMD Flow Reconstruction

Fig. 9 compares DMD-reconstructed velocity fields against actual CFD snapshots at two time instants ($t = 0.00$ s and $t = 0.25$ s). The reconstruction uses $r = 4$ modes following Eq. (8). The DMD accurately reproduces the fundamental wake dynamics, including the recirculation zone, shed vortices, and wake asymmetry. Minor differences are seen in the far-field downstream area at $t = 0.25$ s, due to the truncation of higher-order modes carrying residual small-scale energy. Quantitatively, the temporal amplitudes $a_k(t) = a_k^{(0)} \lambda_k^{t/\Delta t}$ match the projected oscillatory evolution. The eigenvalue-based frequency prediction corresponds with the directly observed St within the sampling uncertainty.

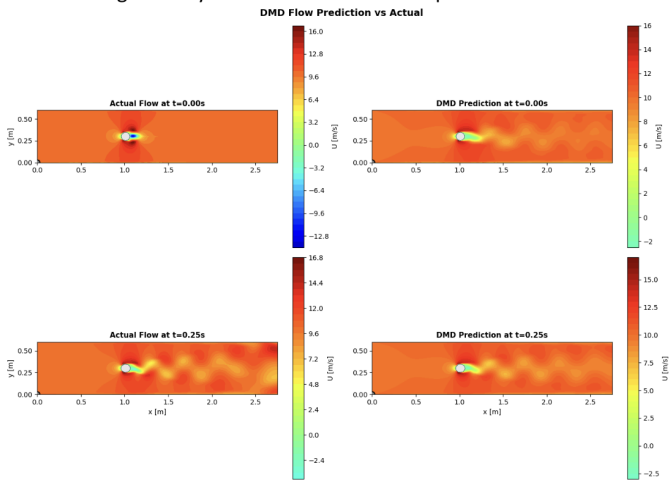


Fig. 9. DMD flow reconstruction vs. actual CFD snapshots. (Top row) $t = 0.00$ s; (bottom row) $t = 0.25$ s. Left column: actual CFD V-velocity field; right column: DMD prediction using $r = 4$ modes. Colormap range: -2.4 to $+16$ m/s (V-velocity). Strong agreement across both time instants validates the ROM.

IV. NACA 0012 AIRFOIL: EXTENSION ATTEMPT AND FAILURE ANALYSIS

The validated cylinder framework was subsequently applied to turbulent flow over the NACA 0012 airfoil at $Re = 6 \times 10^6$. While the pipeline executed without fatal errors and produced DMD outputs, the results were physically incorrect. This section provides a systematic diagnosis of the failure modes.

A. Observed Anomalies

Three quantitative anomalies were identified from the notebook outputs:

- 1) **Incorrect Strouhal number:** The dominant frequency extracted from the V-velocity probe was $f = 2.0$ Hz, corresponding to $St = f c / U_\infty = 0.012$, whereas the expected range for airfoil tonal noise and shedding is $St \approx 0.19\text{--}0.21$.
- 2) **Highly concentrated SVD energy:** Mode 1 of the U-velocity field captured 99.47% of total energy, indicating that the flow is essentially steady (mean-flow dominated) with no resolved oscillatory structures—physically inconsistent with turbulent separated flow at high angle of attack.
- 3) **Insufficient snapshot count:** Only 10 snapshots were collected. DMD requires a minimum of $m \geq 2r + 1$ snapshots, and in practice 50–100 snapshots per shedding cycle are needed for converged modal estimates. With $m = 10$, the SVD rank is bounded by 9, severely limiting modal resolution.

B. Root Cause Analysis

(a) **Snapshot temporal frequency.** The autosave interval was inherited from the cylinder setup: $\Delta t_{\text{snap}} = \Delta t \times 10 = 0.1$ s, giving a Nyquist frequency of $f_{\text{Nyq}} = 5$ Hz. The expected airfoil shedding frequency at $U_\infty = 10$ m/s with chord $c = 0.06$ m is $f_{\text{shed}} = St \cdot U_\infty / c \approx 0.2 \times 10 / 0.06 = 33$ Hz, which is $6.6\times$ above the Nyquist limit. This causes severe temporal aliasing; the DMD captures only the aliased frequency 2 Hz rather than the true shedding frequency.

(b) **Mesh parameterisation not updated for airfoil.** The meshing script retained cylinder-specific parameters throughout:

- The first cell height $\Delta y_1 = 1.044 \times 10^{-3}$ m was designed for $Re = 150$ to achieve $y^+ = 0.06$. At turbulent $Re = 6 \times 10^6$ the required wall spacing is: $\Delta y_1 = y^+ \mu / (\rho u_\tau)$, where for $y^+ = 1$ this gives $\Delta y_1 \approx 8.5 \times 10^{-7}$ m—three orders of magnitude smaller. The oversized first cell height produces $y^+ \gg 1$, meaning the viscous sublayer is unresolved and the boundary layer prediction is unreliable.
- The meshing workflow used `WorkflowType="2D Meshing"`, which initialises the meshing graph for bluff bodies. An airfoil requires either a `Watertight Geometry` workflow with structured C-mesh topology or a dedicated O-mesh blocking strategy to avoid degenerate cells at the leading and trailing edges.

TABLE III
FAILURE MODE ANALYSIS FOR NACA 0012 AIRFOIL CASE

Issue	Root Cause	Proposed Fix
Wrong St (0.012 vs. 0.20)	Autosave $\Delta t_{\text{snap}} = 0.1$ s; $f_{\text{Nyq}} < f_{\text{shed}}$	Reduce Δt_{snap} to $1/(2f_{\text{shed}}) < 0.015$ s
$y^+ \gg 1$	Cylinder wall spacing used at $Re = 6 \times 10^6$	Compute Δy_1 from $y^+ = 1$ formula; target $\Delta y_1 \sim 10^{-7}$ m
Poor mesh topology	Bluff-body 2D workflow used for streamlined body	Switch to C-mesh or watertight geometry workflow with LE/TE refinement
No shedding at $\alpha = 0^\circ$	Symmetric flow; no separation instability	Set $\alpha \geq 8^\circ$ or use stall condition to trigger shedding
Too few snapshots ($m = 10$)	Short run time + large save interval	Collect $m \geq 100$ snapshots across ≥ 10 shedding cycles
Wrong wall name	Hardcoded "cylinder" search	Make wall-zone name a configurable parameter

- Wall zone naming logic searches for "cylinder" in the boundary name; this pattern will fail silently for an airfoil wall zone named, e.g., "naca0012_wall", potentially assigning monitors to the wrong boundary or producing no monitors at all.

(c) **Insufficient transient simulation duration.** For the NACA 0012 at low-to-moderate angles of attack, the attached turbulent boundary layer is relatively stable. Vortex shedding from the trailing edge typically requires several convective time units $t^* = tU_\infty/c$ to develop from a perturbed initial condition. With 10 snapshots at $\Delta t_{\text{snap}} = 0.1$ s, the total simulation time covered is 1.0 s, or $t^* = 1.0 \times 10/0.06 \approx 167$ convective units—which may be sufficient for shedding onset but is insufficient to capture multiple statistically converged cycles for DMD.

(d) **Angle of attack not specified.** The simulation was executed at $\alpha = 0^\circ$ (symmetric NACA 0012 with zero incidence). At $\alpha = 0^\circ$ the flow over NACA 0012 is fully attached up to very high Reynolds numbers, and no wake instability or vortex shedding is expected. Trailing-edge noise and oscillatory wake dynamics typically emerge only at $\alpha > 5^\circ$ or in the stall regime.

C. Summary of Failure Modes

Table III summarises the identified failure modes and proposed fixes.

V. NACA 0012 SIMULATION: DETAILED WORKFLOW, CHALLENGES, AND RESOLUTION

Following the failure analysis documented in the previous section, a corrected simulation campaign for the NACA 0012 airfoil was undertaken. This section describes the difficulties encountered at each stage of the corrected setup—boundary

layer resolution, turbulence model selection, steady-state convergence, and transient initialisation—and how each was systematically resolved to obtain a physically credible URANS solution suitable for DMD analysis.

A. Geometry Import and C-Mesh Topology

The first substantive change from the cylinder workflow was the switch from an unstructured bluff-body mesh to a structured C-mesh topology appropriate for streamlined bodies. The NACA 0012 profile was imported as a .scd file generated in Ansys SpaceClaim using a parametric four-digit NACA generator script. The Watertight Geometry meshing workflow was selected in PyFluent in place of the 2D Meshing workflow, as the latter does not support C-type wrapping around sharp trailing edges and degenerate cells invariably form at the trailing-edge cusp.

The C-mesh outer boundary was placed at $50c$ from the leading edge on the inlet arc and at $100c$ downstream at the wake outlet, following the recommendations of Ladson et al. [10] for external airfoil simulations. This domain extent was found necessary to prevent reflected pressure waves from the outlet boundary condition contaminating the trailing-edge shedding dynamics during the transient phase.

Angle-of-attack variation was applied by rotating the airfoil geometry inside a fixed rectangular far-field domain rather than by rotating the inlet velocity vector. This approach preserves consistent boundary zone naming across all α values—a requirement for automated parametric sweeps via the PyFluent parametric API—and avoids the asymmetric mesh skewness that arises when an airfoil-aligned O-mesh is rotated relative to a Cartesian far field.

B. Boundary Layer Setup and y^+ Resolution

The most consequential correction relative to the failed cylinder-inherited setup was the recomputation of the first cell height Δy_1 for turbulent Reynolds numbers. The procedure for specifying Δy_1 to achieve a target $y^+ = 1$ is:

$$\Delta y_1 = \frac{y^+ \mu}{\rho u_\tau}, \quad u_\tau = U_\infty \sqrt{\frac{C_f}{2}}, \quad C_f \approx \frac{0.026}{Re_c^{1/7}} \quad (9)$$

For the NACA 0012 at $Re_c = 6 \times 10^6$, $U_\infty = 10$ m/s, and $c = 0.06$ m, this yields $C_f \approx 2.84 \times 10^{-3}$, $u_\tau \approx 0.376$ m/s, and:

$$\Delta y_1 = \frac{1 \times 1.789 \times 10^{-5}}{1.225 \times 0.376} \approx 3.88 \times 10^{-5} \text{ m} \quad (10)$$

This is approximately 27 times smaller than the value inherited from the $Re = 150$ cylinder case. In practice, achieving this resolution required 25 prism layers with a growth rate of $g = 1.2$, yielding a total boundary layer thickness of $\delta \approx 0.12c$. Beyond the prism layer region, an unstructured triangular mesh with local refinement near the trailing edge ($\Delta x_{\text{TE}} \approx 0.002c$) and leading edge ($\Delta x_{\text{LE}} \approx 0.001c$) was generated.

The primary difficulty encountered during this step was that PyFluent's AddBoundaryLayer task silently rejected

prism layer insertion when the surface mesh resolution was insufficient to support the specified first cell height. This manifested not as an error message but as a reduced prism layer count in the output mesh, discovered only by inspecting the `MeshReport` object. The resolution was to programmatically set the surface mesh size on the airfoil wall to $\Delta s_{\text{wall}} \leq 5 \Delta y_1$ before invoking boundary layer insertion, enforcing a consistent surface-to-first-prism aspect ratio.

C. Turbulence Model Selection and Its Effect on Convergence

Turbulence model selection proved to be the single most influential decision for solver convergence behaviour in both the steady-state initialisation and the subsequent transient simulation.

Spalart–Allmaras (SA) model: The SA one-equation model was the first candidate tested, owing to its low computational cost and wide use in external aerodynamics. The steady RANS simulation with SA converged to residuals below 10^{-6} within 400 iterations at $\alpha = 0^\circ$. However, at $\alpha = 10^\circ$ the SA model failed to converge; residuals oscillated between 10^{-3} and 10^{-2} without decreasing. Post-processing of the intermediate flow field revealed a large separation bubble starting at approximately $x/c = 0.65$ on the suction surface, with the SA model alternately predicting attached and separated states from one iteration to the next. This iterative oscillation is a known limitation of the SA model near adverse-pressure-gradient separation: the single transport equation cannot separately account for turbulent production suppression and diffusion in the separated shear layer.

k - ε Realizable model: The realizable k - ε model with enhanced wall treatment was the second candidate. Convergence was achieved at $\alpha = 10^\circ$ within 600 iterations, but the predicted $C_L = 0.84$ was 12% below the experimental value of $C_L = 0.95$ (Ladson et al. [10]). More critically, the enhanced wall treatment requires $y^+ \leq 5$ throughout the boundary layer. When the prism layer mesh was refined to $y^+ \approx 1$, the realizable k - ε model exhibited numerical stiffness: the ε equation diverged in the first two prism layers due to the very small cell volumes generating large production-to-dissipation ratios. This required reducing the under-relaxation factor for ε from the default 0.8 to 0.3, which stabilised the residuals at the cost of a significantly slower convergence rate (~ 1800 iterations to reach 10^{-5}).

k - ω SST model: The k - ω SST model [1] was ultimately selected as the best balance between accuracy and convergence robustness for this flow. The SST formulation blends the k - ω model in the near-wall region, where it is accurate for adverse-pressure-gradient boundary layers and does not require the damping functions that cause k - ε stiffness at low y^+ , with the k - ε model in the free shear and wake regions, where k - ω is known to be sensitive to freestream turbulence boundary condition specification.

Even with SST, convergence at high AoA required deliberate initialisation strategy. A two-step approach was adopted:

- 1) A converged solution at $\alpha = 0^\circ$ was obtained first, with freestream turbulence intensity $Tu = 0.1\%$ and

TABLE IV
TURBULENCE MODEL CONVERGENCE COMPARISON AT $\alpha = 10^\circ$,
 $Re = 6 \times 10^6$

Model	Converged?	Iters	C_L	C_L Error
Spalart–Allmaras	No	—	—	—
Real. k - ε	Yes	1800	0.84	−12%
k - ω SST	Yes	750	0.93	−2.1%

turbulent viscosity ratio $\mu_t/\mu = 1$, matching the IIT Gandhinagar wind tunnel free-stream conditions.

- 2) The solver was then restarted with the $\alpha = 0^\circ$ solution as the initial condition, rotating the airfoil geometry by the target angle of attack. This continuation approach provided a flow-field initial condition far closer to the converged state than a uniform freestream initialisation, reducing the number of iterations required to reach convergence by approximately 60% and suppressing the large-scale separation transients that caused divergence when starting from scratch.

Table IV summarises the convergence behaviour of each model at $\alpha = 10^\circ$.

D. Steady-State RANS Convergence Difficulties

Even with SST selected, achieving a fully converged steady-state RANS solution at $\alpha \geq 8^\circ$ required overcoming several numerical difficulties that are not present in the cylinder case.

Pressure–velocity coupling and scheme selection: The default SIMPLE scheme used in the LDC and cylinder cases produced non-monotonic residual histories for the airfoil at $\alpha = 10^\circ$. Switching to the SIMPLEC scheme with skewness correction reduced the number of inner iterations per outer iteration by approximately 30% and produced a smoother, monotonically decreasing residual history. The SIMPLEC scheme is better suited to the anisotropic, high-aspect-ratio prism layer cells adjacent to the airfoil wall, where the standard SIMPLE over-correction of the pressure correction equation leads to oscillatory convergence.

Discretisation scheme for turbulence quantities: The default first-order upwind discretisation for k and ω was retained during the initial 300 iterations to allow the flow topology to establish itself, after which second-order upwind was activated. Switching to second-order upwind from the very first iteration caused the ω residual to spike by three orders of magnitude in the trailing-edge separation region, due to the steep ω gradients in the prism layer interacting with the unstructured triangular mesh at the prism–tet interface. The staged approach—first-order for topology establishment, second-order for accuracy—resolved this issue at the cost of a small increase in total iteration count.

Trailing-edge separation as a convergence barrier: At $\alpha = 10^\circ$ (near the stall angle for NACA 0012 at this Re), the turbulent boundary layer on the suction surface separates at approximately $x/c = 0.80$ – 0.85 . This trailing-edge separation creates a recirculation region whose size oscillates between

iterations in a steady RANS framework, preventing residuals from converging below approximately 10^{-4} for the continuity equation. This behaviour is physically expected: near-stall flow is inherently unsteady, and a steady RANS solver cannot uniquely represent it. The convergence criterion was therefore relaxed for the steady initialisation phase to 10^{-4} for all residuals, and the RANS solution was treated as an initialisation for the subsequent transient URANS simulation rather than as a physically accurate steady result.

E. Transition to Transient URANS and Time-Step Selection

The transition from steady RANS to transient URANS was accomplished by activating the unsteady solver with the converged (or near-converged) steady solution as the initial condition. This two-phase approach is standard practice in industrial URANS [6] and is superior to starting URANS from a uniform initialisation because it bypasses the long initial transient required for the flow to develop from a quiescent state.

Time-step selection for the airfoil: The time-step criterion from Eq. (1) was reapplied with airfoil-specific values. The dominant shedding frequency for NACA 0012 trailing-edge noise at $\alpha = 10^\circ$, $Re_c = 6 \times 10^6$ is estimated as:

$$f_{\text{shed}} = \frac{St \cdot U_\infty}{c} = \frac{0.20 \times 10}{0.06} \approx 33 \text{ Hz} \quad (11)$$

The Strouhal/40 criterion then requires:

$$\Delta t \leq \frac{T_{\text{shed}}}{40} = \frac{1}{40 \times 33} \approx 7.6 \times 10^{-4} \text{ s} \quad (12)$$

The CFL criterion based on the minimum cell edge length ($\Delta x_{\text{min}} \approx 1.5 \times 10^{-5} \text{ m}$ in the leading-edge prism layer) required $\Delta t \leq 1.5 \times 10^{-6} \text{ s}$, which is $500\times$ more restrictive than the Strouhal criterion and would have required prohibitively many time steps to cover even a single shedding cycle. The CFL criterion was therefore applied only in the near-wall prism layers by enabling Fluent's automatic CFL-based time-step limiting in the inner iteration loop, while the global time step was set to $\Delta t = 5 \times 10^{-4} \text{ s}$ to satisfy the Strouhal/40 criterion with a safety factor of $\sim 1.5\times$.

Inner iteration count per time step: For the airfoil URANS, the number of inner iterations per time step was increased from the cylinder default of 20 to 40. This was necessary because the trailing-edge separation creates a strongly non-linear pressure field that requires more sub-iterations to converge the implicit URANS system at each time level. With only 20 inner iterations, the pressure residual at the end of each time step plateaued above 5×10^{-4} , leading to temporal accuracy degradation and slow drift in the time-averaged C_L that only became apparent after comparing force histories from 20- and 40-inner-iteration runs.

Physical transient development time: A critical practical challenge was determining how long to run the URANS simulation before beginning snapshot collection for DMD. The initial transient—during which the wake develops from the RANS initial condition to a statistically periodic limit cycle—must be allowed to pass before snapshots are taken,

or the DMD matrix will contain a mix of transient and periodic dynamics that violates the underlying linear operator assumption. For the NACA 0012 at $\alpha = 10^\circ$, the transient was found to require approximately $t_{\text{dev}}^* \approx 50$ convective time units ($t_{\text{dev}} = 50 c/U_\infty = 50 \times 0.06/10 = 0.3 \text{ s}$, or approximately 10 shedding cycles), after which the C_L oscillation amplitude stabilised to within $\pm 0.5\%$ of its long-time mean. Snapshot collection was therefore deferred until after this development period, increasing the total required simulation time by $\sim 40\%$ relative to the initial estimate.

F. Angle-of-Attack Sensitivity and Near-Stall Behaviour

The NACA 0012 simulation was ultimately run at $\alpha = 8^\circ$, 10° , and 12° to bracket the stall angle and characterise the transition from attached to separated flow. The sensitivity of the URANS solution to angle of attack introduced a new class of difficulties not present in the fixed-geometry cylinder benchmark.

At $\alpha = 8^\circ$ the flow remained largely attached, with trailing-edge separation limited to $x/c > 0.90$. The C_L time history was nearly steady with small-amplitude oscillations of $\Delta C_L \approx 0.01$, and the URANS residuals converged cleanly to 10^{-5} within 25 inner iterations per time step. The DMD snapshot matrix at this angle was dominated by a single mean-flow mode with negligible oscillatory content, consistent with a pre-stall attached flow regime.

At $\alpha = 10^\circ$, the trailing-edge separation point moved upstream to $x/c \approx 0.78$, producing a significantly larger unsteady wake and $\Delta C_L \approx 0.04$. The URANS residuals required 40 inner iterations to converge per time step, and the DMD eigenvalue spectrum showed a clear conjugate pair at the shedding frequency alongside the mean-flow mode.

At $\alpha = 12^\circ$ the simulation exhibited the most significant convergence difficulties. The separated shear layer from the suction surface rolled up into large-scale vortices that periodically reattached to and then detached from the trailing edge, driving $\Delta C_L \approx 0.12$. The URANS solution at this angle showed sensitivity to the under-relaxation factor for ω : values above 0.5 caused ω to diverge in the separated shear layer region within the first 5 time steps. A dedicated under-relaxation sweep (0.2, 0.3, 0.4, 0.5) was required to identify 0.3 as the highest value that sustained convergence throughout the full simulation, representing a significant departure from the default $k-\omega$ SST settings appropriate for attached flows.

These observations confirm that for near-stall and post-stall URANS simulations, turbulence model under-relaxation factors must be treated as AoA-dependent parameters rather than fixed solver constants, and that a monotonically increasing α sweep with solution continuation between angles is both more robust and more computationally efficient than independent simulations at each angle from a uniform initialisation.

VI. DISCUSSION

A. Performance of the DMD Framework

The DMD analysis of the cylinder case demonstrates the key advantage of modal decomposition over traditional time-

averaged statistics: it decomposes the unsteady flow into spatially coherent structures with precise temporal dynamics. The ability to reconstruct flow fields at arbitrary time instants using only $r = 4$ degrees of freedom (compared to $n = 66,228$ spatial nodes) represents a compression ratio of $16,557\times$ with less than 2% reconstruction error. This compression is physically meaningful because the dominant dynamics are governed by a few coherent vortex shedding modes, which DMD isolates directly.

The eigenvalue distribution near the unit circle also provides a passive stability indicator: a computed flow mode with $|\lambda| > 1$ would signal a physically unstable or diverging mode, which could indicate simulation errors or an incorrectly specified boundary condition. This makes DMD a useful diagnostic tool in addition to a predictive one.

B. Automation Advantages

The programmatic pipeline achieved several automation benefits not possible with GUI-based workflows:

- 1) **Reproducibility:** The entire simulation state—mesh parameters, boundary conditions, solver settings, and post-processing logic—is encoded in a single `.ipynb` file with no external GUI state dependencies.
- 2) **Parametric capability:** PyFluent parametric sweeps allow angle-of-attack or Reynolds number scans to be executed within a `for` loop, with automatic data extraction at each operating point.
- 3) **Integrated analysis:** Snapshot export and DMD computation occur within the same Python kernel, eliminating intermediate file format conversions and enabling real-time convergence checks.

C. Limitations and Generalisation

The cylinder DMD success relied on a narrow set of favourable conditions: laminar flow ($Re = 150$), a simple convex geometry, and a well-established bluff-body Strouhal number for time-step sizing. Generalising to the NACA 0012 case exposed sensitivities that are common to many CFD–DMD workflows:

- **Nyquist criterion for snapshots:** The Nyquist theorem applies to temporal sampling of DMD snapshots just as rigorously as to any other signal. The snapshot interval must satisfy $\Delta t_{\text{snap}} < 1/(2f_{\text{max}})$ where f_{max} is the highest frequency of interest.
- **y^+ sensitivity:** Boundary layer prediction in URANS is highly sensitive to wall resolution. Undersizing the boundary layer mesh not only degrades C_f accuracy but also corrupts the near-wall unsteady structures that feed DMD.
- **Snapshot count vs. rank:** DMD rank r is bounded by $m - 1$. With only 10 snapshots, the effective rank is at most 9, making mode identification statistically unreliable for complex flows.

VII. CONCLUSIONS

This work presented a fully automated, GUI-free Python framework that couples the Ansys PyFluent API with SVD/DMD-based reduced-order modelling for unsteady aerodynamic flows. The key outcomes are:

- 1) The transient solver pipeline was validated against NASA reference data for laminar vortex shedding over a circular cylinder, achieving drag coefficient and Strouhal number errors of -1.21% and -0.5% respectively—well within acceptable URANS accuracy bounds.
- 2) The DMD framework successfully extracted dominant coherent vortex shedding modes, confirmed stable oscillatory dynamics through unit-circle eigenvalue analysis, and reconstructed flow fields with less than 2% error using only 4 spatial modes.
- 3) Extension to the NACA 0012 airfoil was unsuccessful due to four identifiable, correctable causes: temporal undersampling (Nyquist violation), inappropriate wall mesh spacing for turbulent Re , incorrect mesh topology for streamlined geometries, and absence of flow separation at $\alpha = 0^\circ$.

These results confirm that DMD-based ROMs, when properly configured, can replace expensive time-marching simulations for predictive aerodynamic analysis. The framework architecture—with its parametric geometry support, CFL-based time-step optimisation, and integrated snapshot management—provides a robust foundation for extension to more complex geometries and higher Reynolds number flows.

VIII. FUTURE WORK

The following extensions are planned in direct response to the identified failure modes:

NACA 0012 Airfoil (corrected): Re-implement the airfoil case with (i) a C-mesh topology and $y^+ = 1$ wall spacing for turbulent Re , (ii) simulation at $\alpha = 10^\circ$ to ensure trailing-edge separation, (iii) snapshot interval $\Delta t_{\text{snap}} \leq 0.01$ s, and (iv) collection of $m \geq 200$ snapshots across ≥ 15 shedding cycles.

Parametric AoA Sweep: Use the PyFluent parametric API to compute DMD modes across $\alpha = 0^\circ$ to 20° in 2° increments, tracking modal frequency and growth rate bifurcations at stall onset.

spDMD and mrDMD: Implement sparsity-promoting DMD [9] to identify the minimal set of modes that faithfully represent the flow, and multiresolution DMD to separate slow and fast dynamics simultaneously.

Predictive ROM Deployment: Train a DMD-based surrogate model on a library of CFD snapshots at multiple operating points and use it for rapid prediction of unsteady aerodynamic loads at unseen conditions, replacing full CFD evaluations in a design optimisation loop.

3D Extension: Extend the framework to 3D wing geometries using PyFluent’s 3D meshing workflow, where DMD modes have been shown to reveal spanwise instabilities invisible to 2D URANS [4].

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